UNIT 2 · TOPIC 10(B)

Machine Learning and AI

Syllabus topic: Machine learning, IoT, robotics and mechatronics in farm machinery

Machine learning: algorithms that learn patterns from data instead of being explicitly programmed.

Learning type	Data / example
Supervised	Labelled data — classification (weed vs crop), regression (yield prediction)
Unsupervised	Unlabelled data — clustering management zones
Reinforcement	An agent learns actions by reward — robot navigation, spray optimisation

Deep learning

Multi-layer neural networks. **Convolutional neural networks (CNN)** dominate image tasks:

- Weed detection and spot spraying (**"See & Spray"** — herbicide only on weeds, **60–90%** chemical saving).
- Disease/pest identification from leaf images.
- Fruit detection and grading.
- Obstacle detection for autonomous machines.

Other FMP applications

Predictive maintenance from vibration/temperature data, draft/fuel prediction, yield-map cleaning, drone-image analytics (crop stress via NDVI).

$$\mathrm{NDVI} = (\mathrm{NIR} - \mathrm{R})/(\mathrm{NIR} + \mathrm{R})$$

- Healthy vegetation reflects NIR strongly and absorbs red → NDVI near **+0.6–0.9**; a stressed crop reads lower (basis of drone crop monitoring, Unit 10).

Practice MCQs

Q196. Supervised machine learning requires:

(a) Unlabelled data (b) Labelled training data
(c) No data (d) Only images

Q197. The neural-network type dominating image-based weed detection is:

(a) RNN (b) CNN (convolutional)
(c) Perceptron only (d) SOM

Q198. 'See & Spray' technology saves chemicals by:

(a) Diluting spray (b) Spraying only where ML vision detects weeds
(c) Spraying at night (d) Using more nozzles

Q200. NDVI is computed as:

(a) (R−NIR)/(R+NIR) (b) (NIR−R)/(NIR+R)
(c) NIR×R (d) NIR/G

Answers

Q196 — (b) Labelled training data

Q197 — (b) CNN (convolutional)

Q198 — (b) Spraying only where ML vision detects weeds

Q200 — (b) (NIR−R)/(NIR+R) — *Healthy vegetation reflects NIR strongly and absorbs red → NDVI near +0.6–0.9; stressed crop lower (basis of drone crop monitoring, Unit 10).*

Sources: Kepner et al. (hydraulic principles); Drone Rules 2021 (Ministry of Civil Aviation); MoA&FW SOPs for drone application of pesticides & nutrients; SMAM drone guidelines & PIB releases (Kisan Drone, Namo Drone Didi — verified July 2026); ISO 11783 (ISOBUS); NPTEL lectures on mechatronics & precision agriculture.